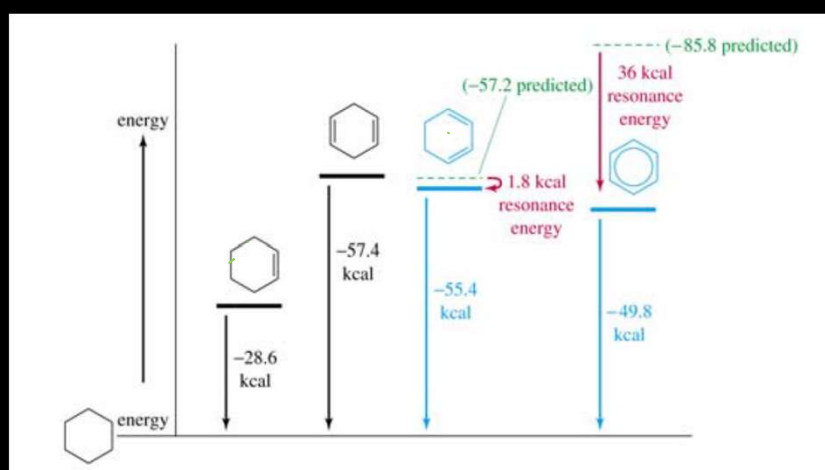
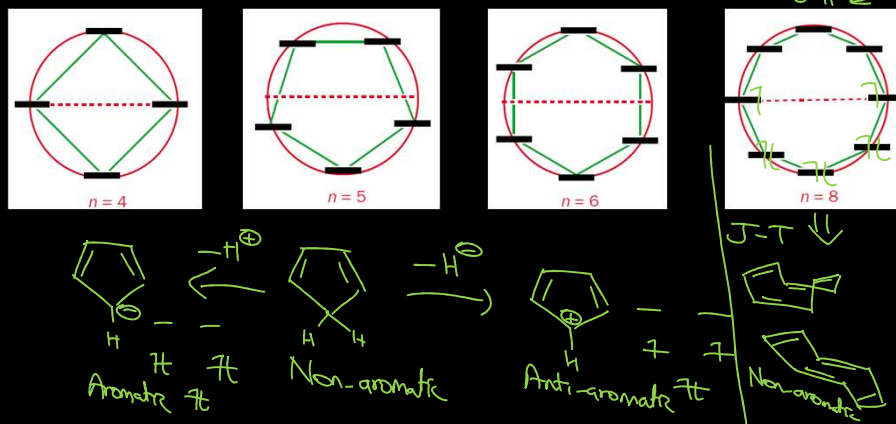


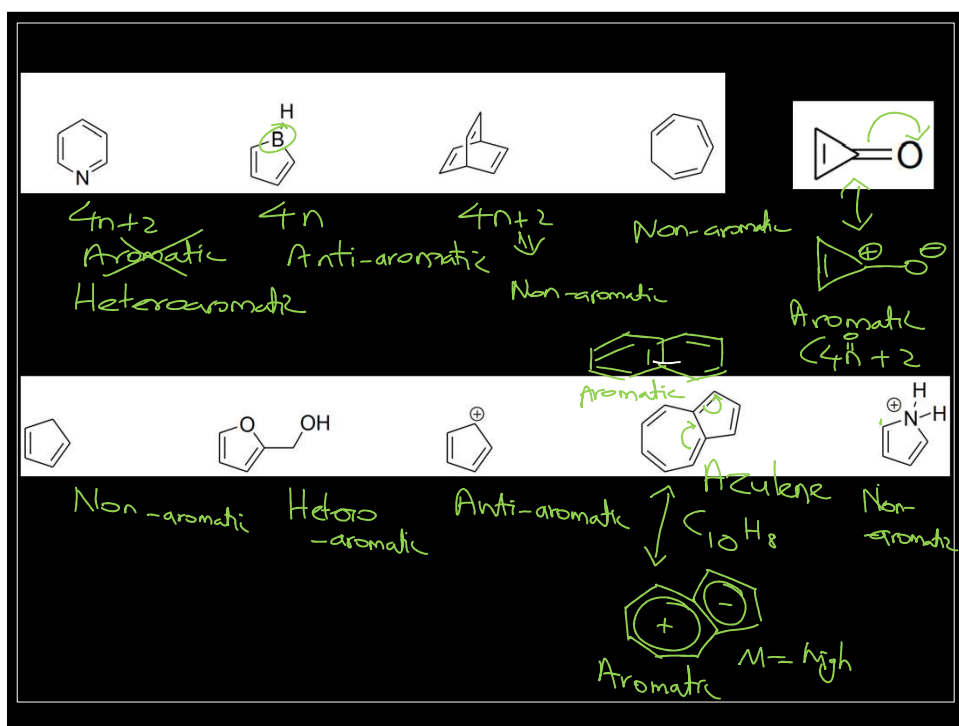
Aromaticity

Hückel's rule

Planar, fully conjugated, monocyclic systems with $(4n + 2)$ – electrons have a closed shell of electrons all in bonding orbitals and are exceptionally stable. Such systems are said to be aromatic.

Analogous systems with $4n$ – electrons are described as anti-aromatic





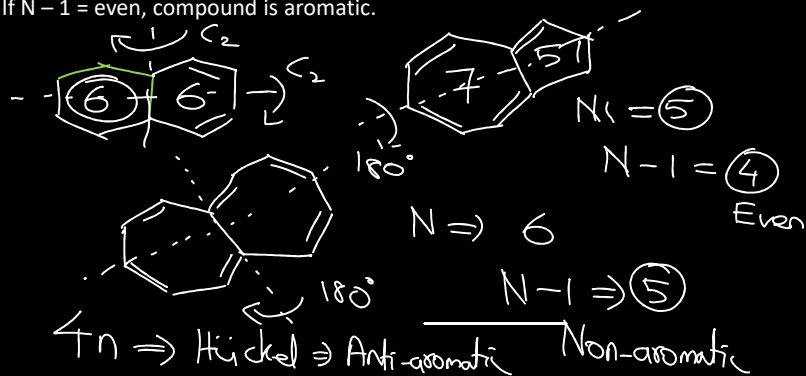
Craig's Rule of aromaticity:

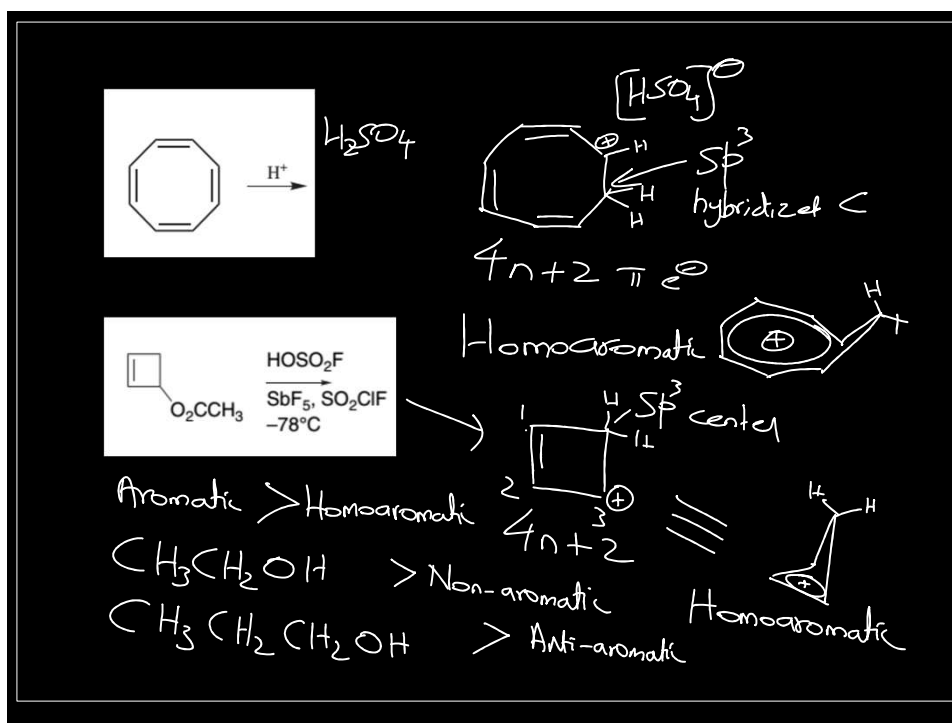
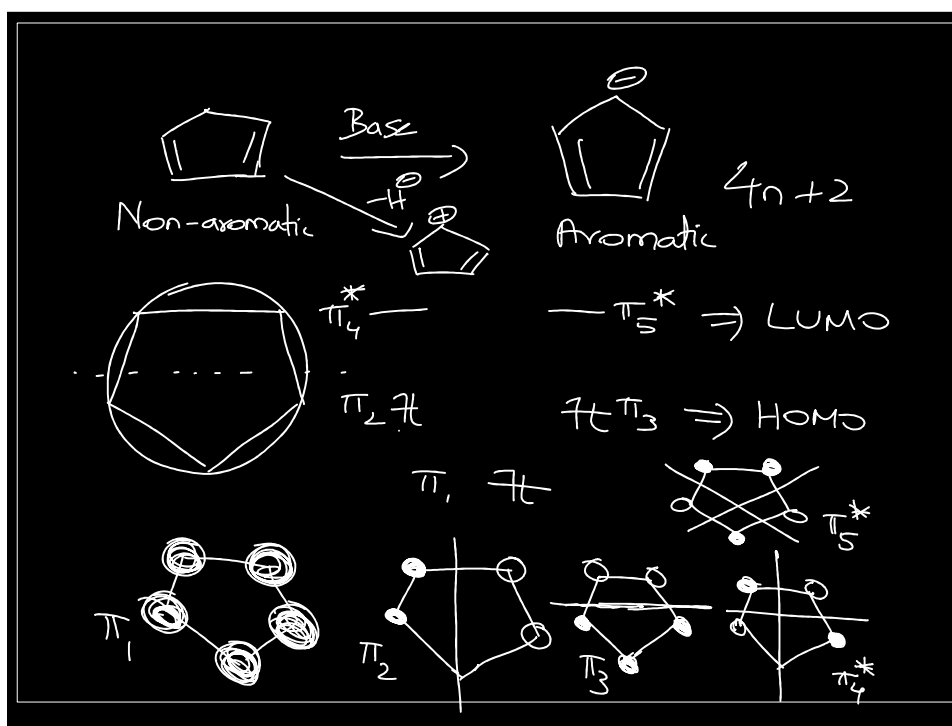
Applicable for polycyclic non-benzenoid compounds.

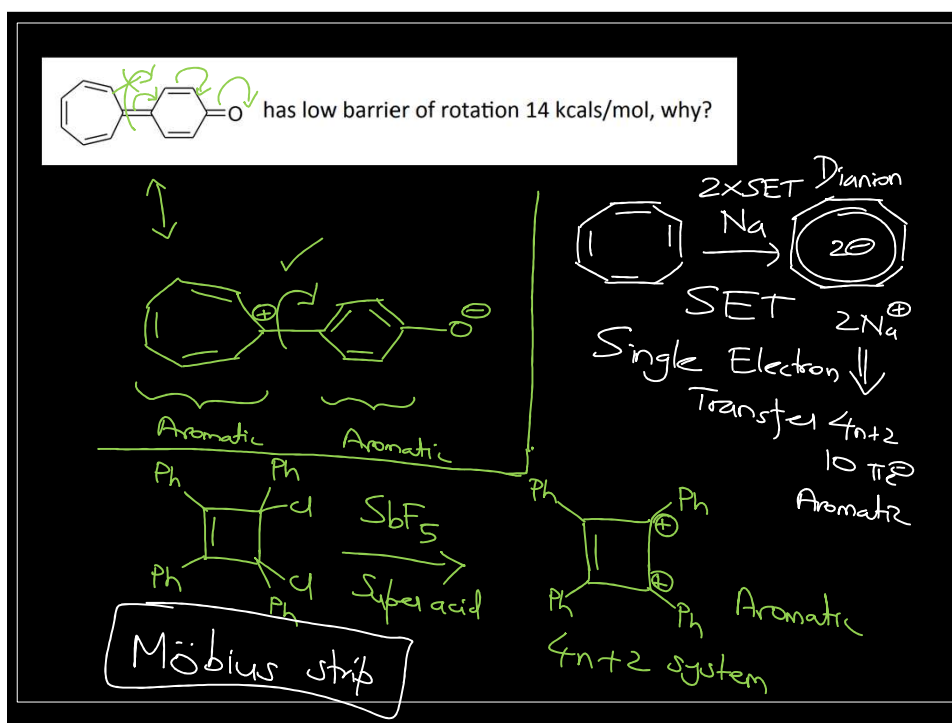
If molecule contain C_2 -axis then count total number of double bonds (N) and calculate value of $N-1$, which can be useful for deciding aromaticity characteristics.

(1) If $N-1 = \text{odd}$, compound is non-aromatic

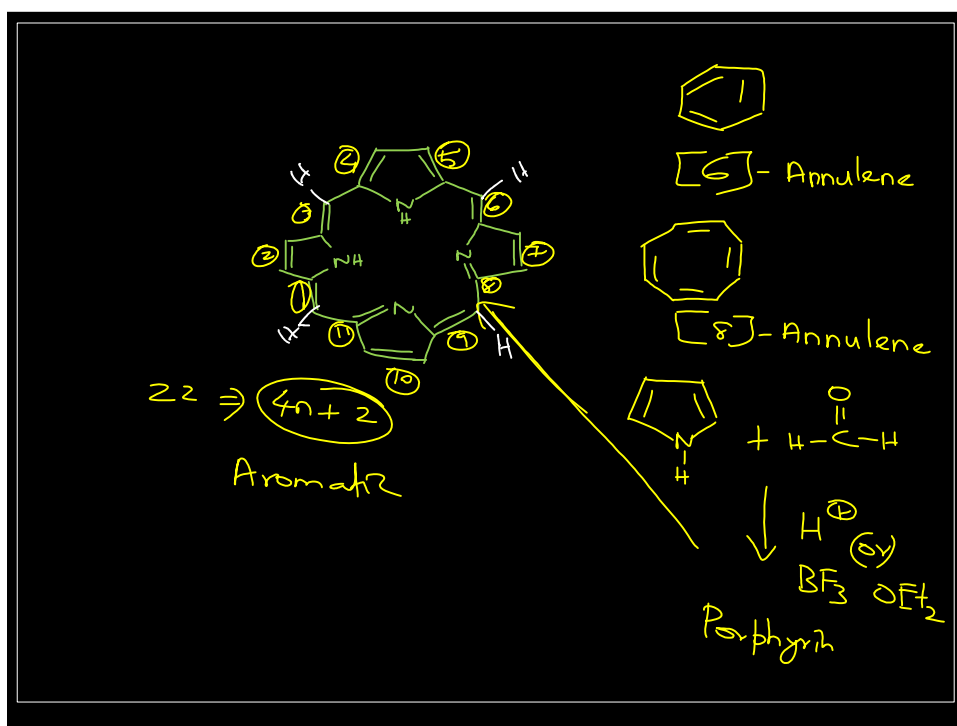
(2) If $N-1 = \text{even}$, compound is aromatic.







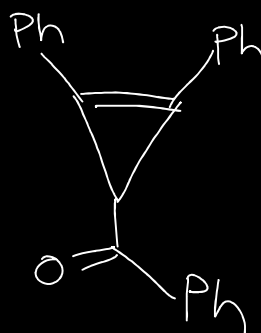
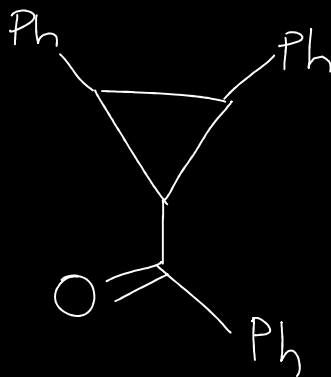
Hückel		Möbius
Aromatic	$4n+2$ (No node) Even nodes	$4n$ One node Odd nodes
Antiaromatic	$4n$	$4n+2$



Which of the alpha protons can be deprotonated under base conditions?



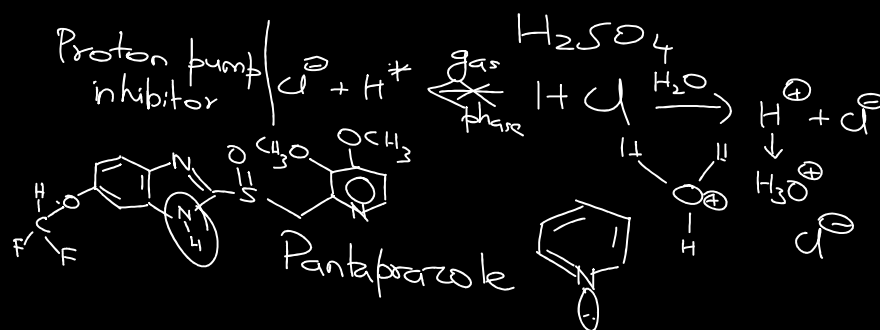
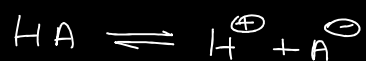
9.05



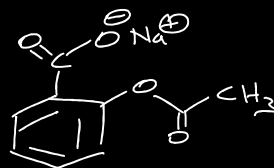
Acids/Bases

pK_a

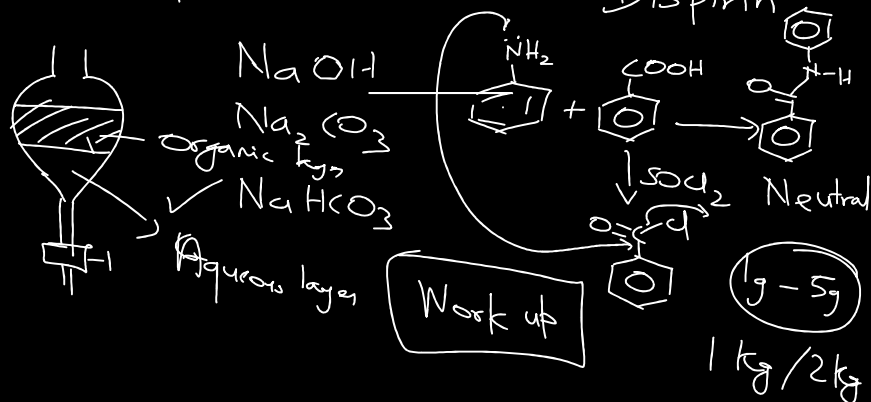
$pH \Rightarrow [H_3O^+]$

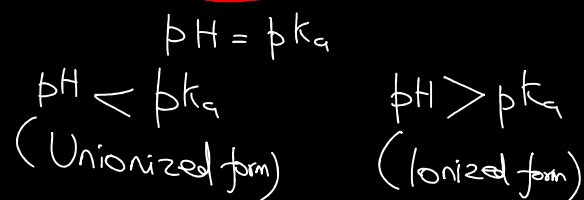


Base



Dispirin





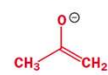
7

① Size of anion

② Beta leaving group
 $\downarrow$
 Bond strength

③ Stability of the conjugate base

$\text{HA} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{A}^-$
 Acid Base Conjugate acid Conjugate base

Acid	$\text{p}K_a$	Conjugate base
HI	ca. -10	I^-
HCl	ca. -7	Cl^-
H_2SO_4	ca. -3	HSO_4^-
HSO_4^-	2.0	SO_4^{2-}
CH_3COOH	4.8	CH_3COO^-
H_2S	7.0	HS^-
NH_4^+	9.2	NH_3
$\text{C}_6\text{H}_5\text{OH}$	10.0	$\text{C}_6\text{H}_5\text{O}^-$
CH_3OH	15.5	CH_3O^-
	20.0	
	25	
NH_3	33	NH_2^-
C_6H_6	ca. 43	C_6H_5^-
CH_4	ca. 48	CH_3^-

A Brønsted acid is defined as a proton donor

A Brønsted base is a proton acceptor

A Lewis acid is an electron pair acceptor

A Lewis base is an electron pair donor

$$\text{HA} + \text{H}_2\text{O} \xrightleftharpoons{K_a} \text{H}_3\text{O}^+ + \text{A}^-$$

$$\text{p}K_a = -\log K_a \quad K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{AH}]}$$

$$\begin{array}{ccccccc} \text{HA} & + & \text{B} & \rightleftharpoons & \text{A}^- & + & \text{BH}^+ \\ \text{Acid} & & \text{Base} & & \text{Conjugate base} & & \text{Conjugate acid} \end{array}$$

The stronger the acid HA, the weaker its conjugate base A^- .

The stronger the base A^- , the weaker its conjugate acid AH.

The most important factor in the strength of an acid is the stability of the conjugate base—the more stable the conjugate base, the stronger the acid.

An important factor in the stability of the conjugate base is which element the negative charge is on—the more electronegative the element, the more stable the conjugate base.

HA vs HI

← Bond strength increases

Hand-drawn chemical structures illustrating bond strength trends, including $\text{HO-SO}_3\text{H}$, CH_3COOH , $\text{C}_6\text{H}_5\text{OH}$, and CH_3OH .

Acid	pK_a	Conjugate base
HI	ca. -10	I^-
HCl	ca. -7	Cl^-
H_2SO_4	ca. -3	HSO_4^-
HSO_4^-	2.0	SO_4^{2-}
CH_3COOH	4.8	CH_3COO^-
H_2S	7.0	HS^-
NH_4^+	9.2	NH_3
$\text{C}_6\text{H}_5\text{OH}$	10.0	$\text{C}_6\text{H}_5\text{O}^-$
CH_3OH	15.5	CH_3O^-
CH_3COCH_3	20.0	$\text{CH}_3\text{C}(\text{O})\text{CH}_2^-$
$\text{CH}\equiv\text{C-H}$	25	$\text{CH}\equiv\text{C}^-$
NH_3	33	NH_2^-
C_6H_6	ca. 43	C_6H_5^-
CH_4	ca. 48	CH_3^-

Acid **Conjugate base** **pK_a**

methane CH_4	CH_3^-	~48
ammonia NH_3	amide ion NH_2^-	~33
water H_2O	hydroxide ion HO^-	~16
HF	fluoride ion F^-	3

Handwritten notes: $\text{EN} \sim 48$, $\text{element} \sim 33$, $\text{bearing} \sim 16$, $\text{charge} \sim 3$.

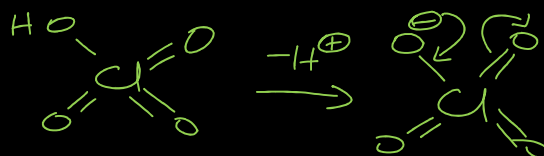
Base / Sodamide $\text{Na}^+ \text{NH}_2^-$

Acid	Conjugate base	pK_a
HF	fluoride ion F^-	3
HCl	chloride ion Cl^-	-7
HBr	bromide ion Br^-	-9
HI	iodide ion I^-	-10

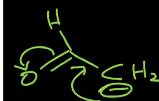
Handwritten notes: **Bond strength**, **Umpolung (Charge inversion)**, $\text{CH}_3\text{-Cl} \xrightarrow{2\text{Li}} \text{CH}_3\text{Li} + \text{LiClMg}$, **LDA**, **Lithium Diisopropyl amide**, $\text{Y-N}^-\text{Li}^+$, -78°C .

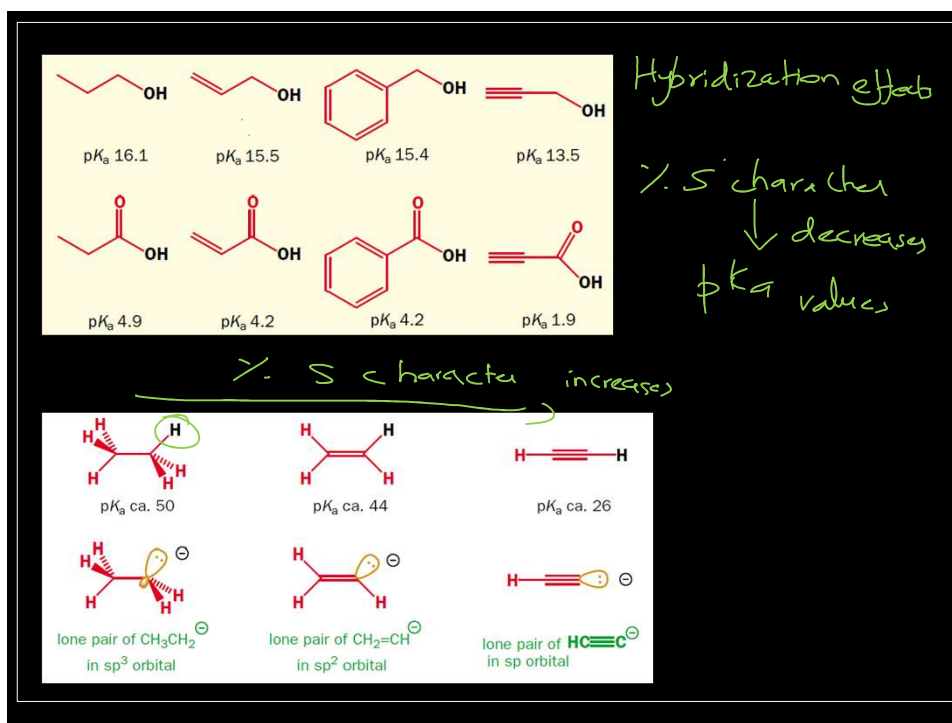
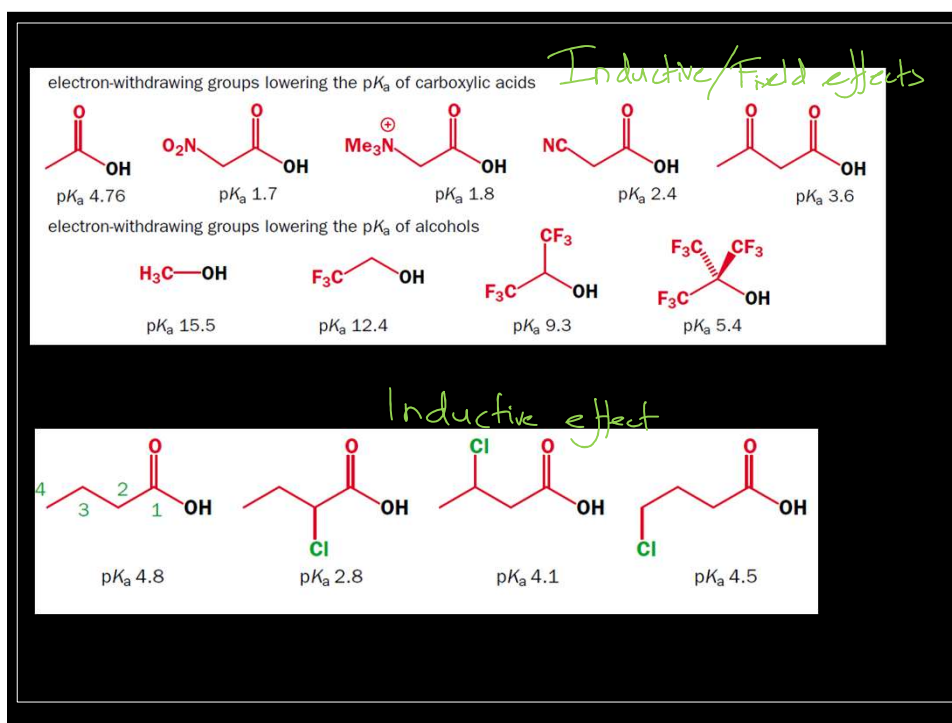
Acid	Conjugate base	pK _a
hypochlorous acid HO-Cl	ClO ⁻	7.5
chlorous acid HO-ClO	ClO ₂ ⁻	2
chloric acid HO-ClO ₂	ClO ₃ ⁻	-1
perchloric acid HO-ClO ₃	ClO ₄ ⁻	-10

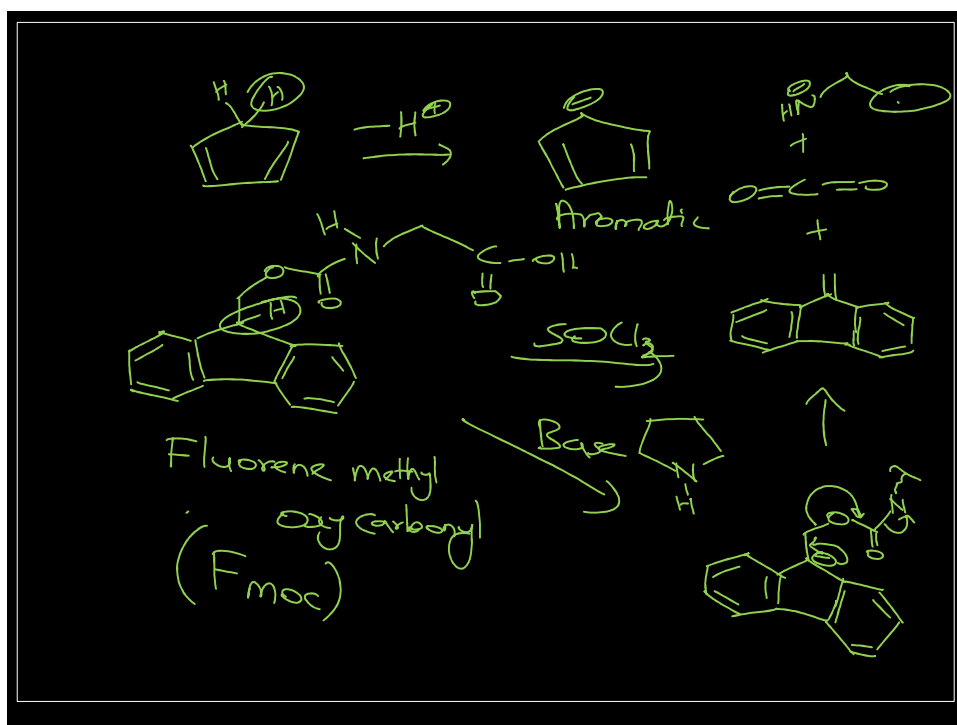
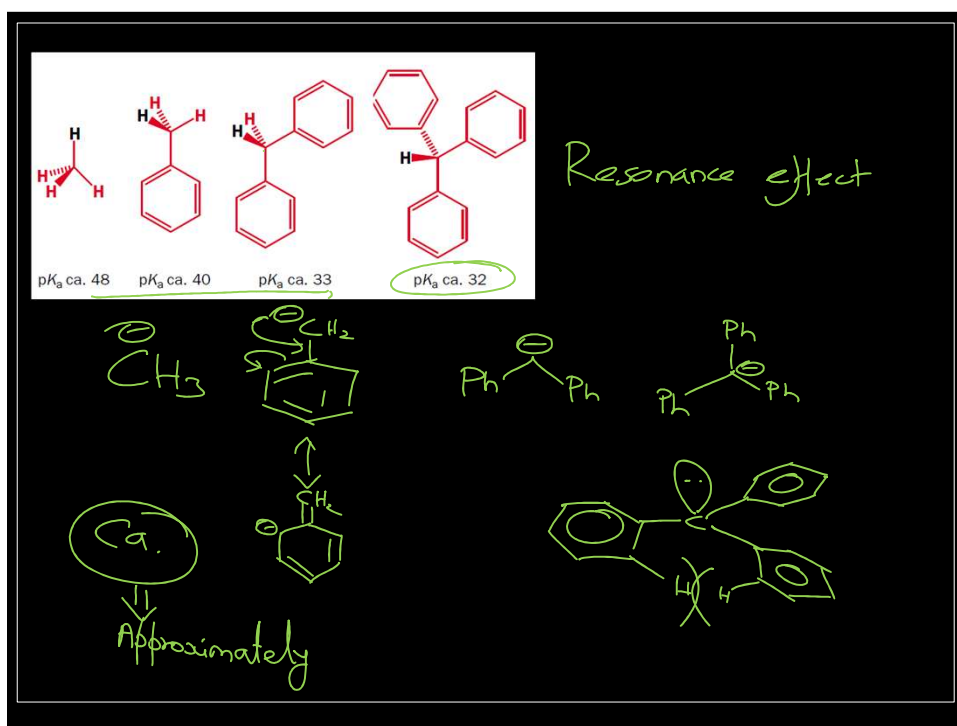
Resonance



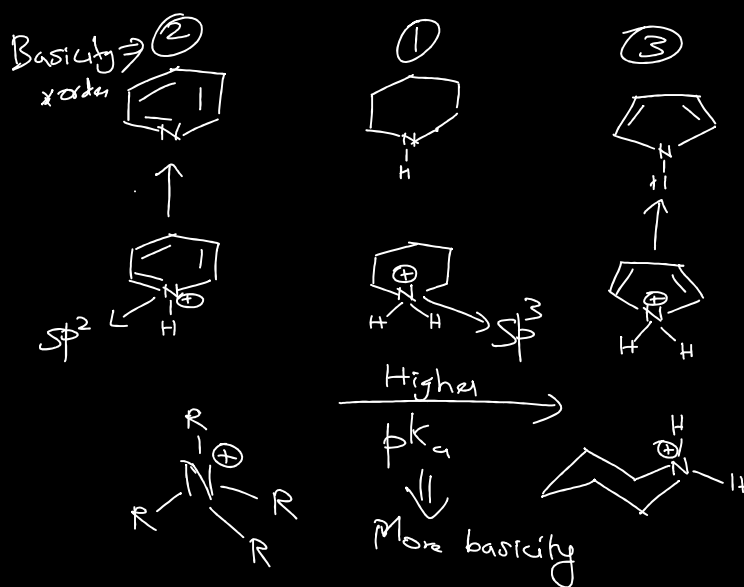
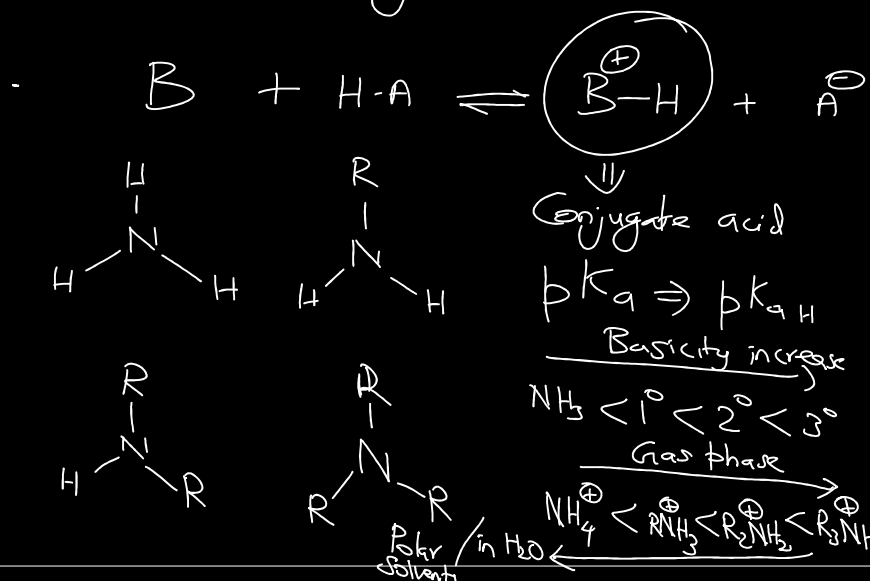
Acid	Conjugate base	pK _a	Comments
		~50	charge is localized on one carbon—difficult since carbon is not very electronegative
		~43	charge is delocalized over π system—better but still not really good
		13.5	charge is delocalized over π system but is mainly on the electronegative oxygen—much better
		5	charge delocalized over π system but mainly over two oxygens—better still
		~48	charge is localized on one carbon—again very unsatisfactory
		10	charge is delocalized but mainly on oxygens of nitro group
		4	charge can be delocalized over two nitro groups—more stable anion
		0	charge can be delocalized over three nitro groups—very stable anion

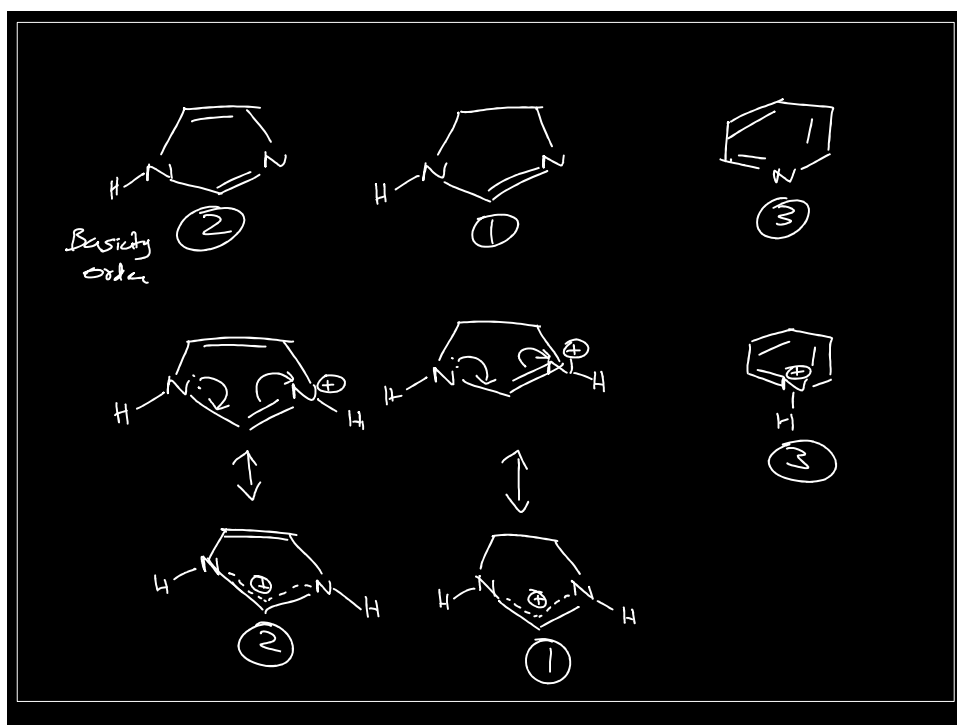
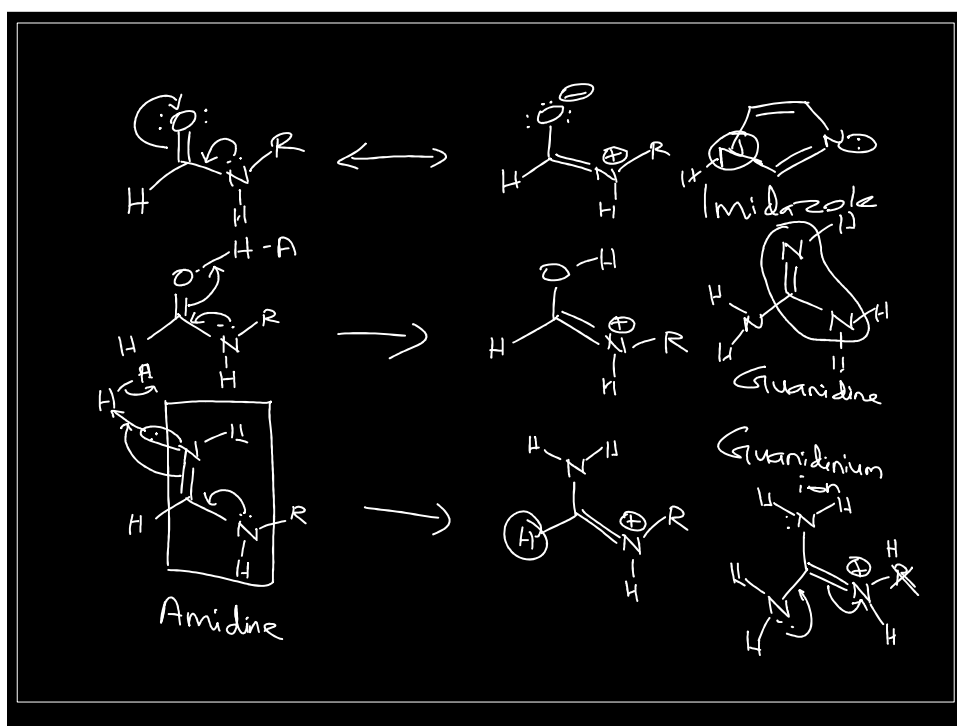


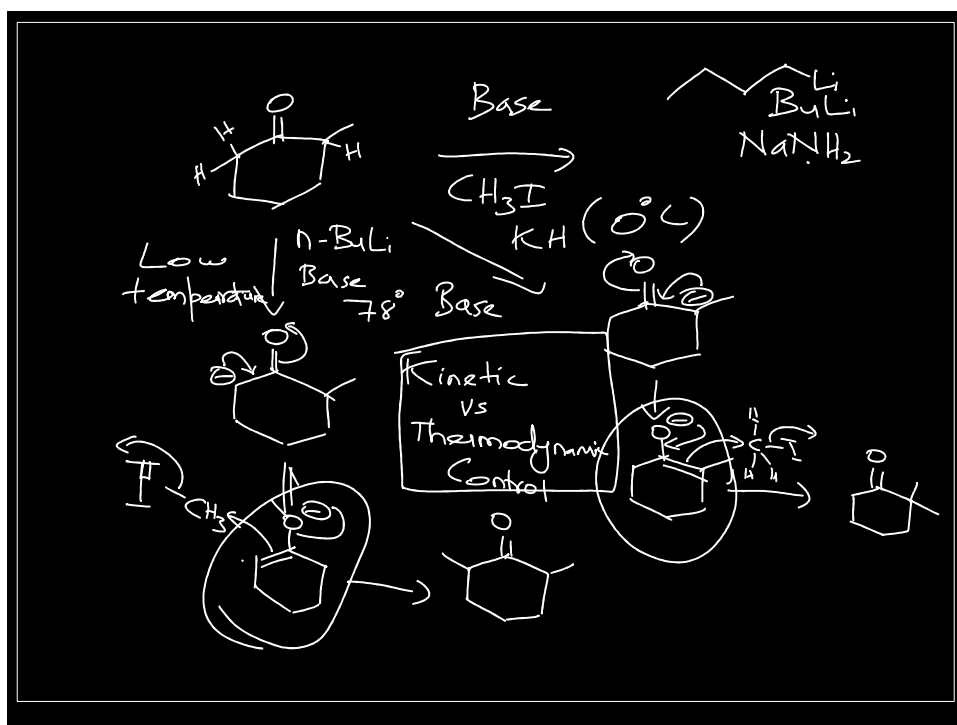
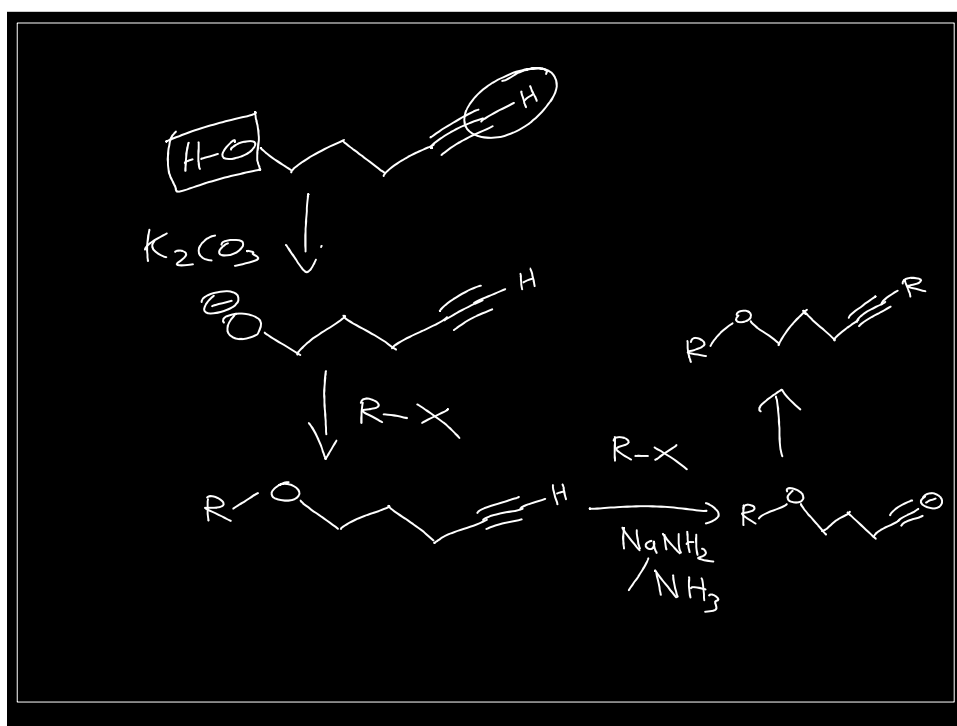




Basicity





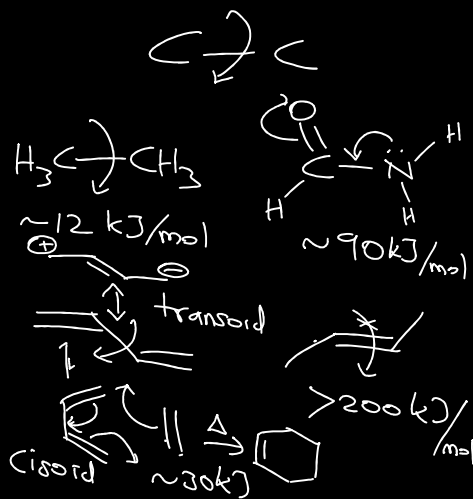


"Life is like riding a bicycle. To keep your balance, you must keep moving."

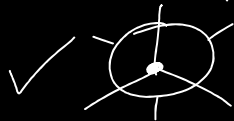
Albert Einstein's life advice in a letter to his son Eduard on 5 February 1930. In the picture, Einstein is riding a bicycle in Santa Barbara, USA in 1933.



Conformations



Newman projection



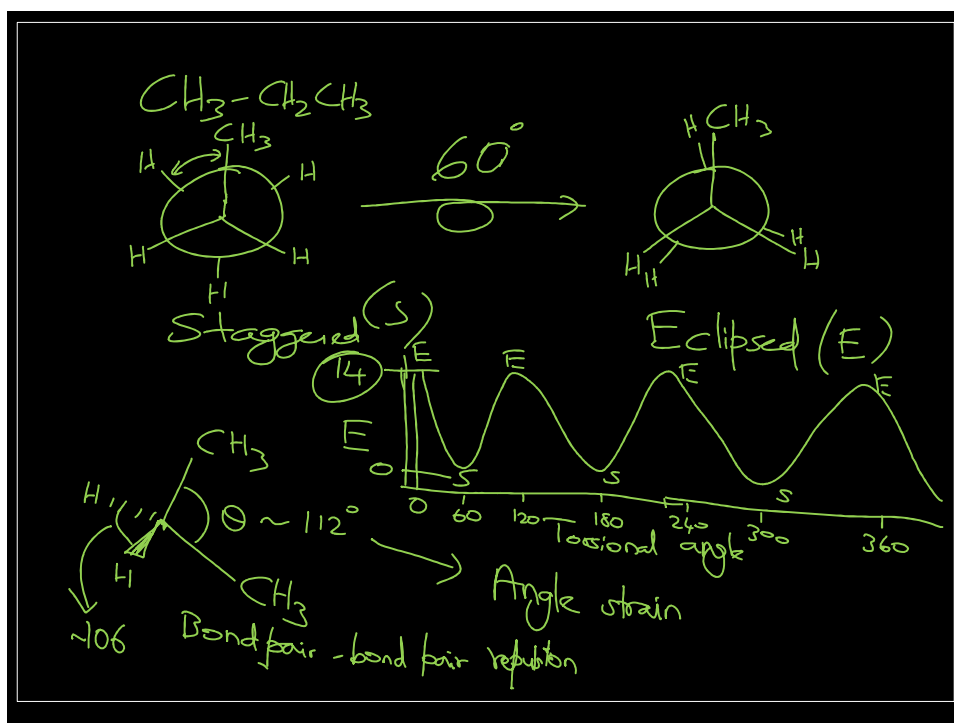
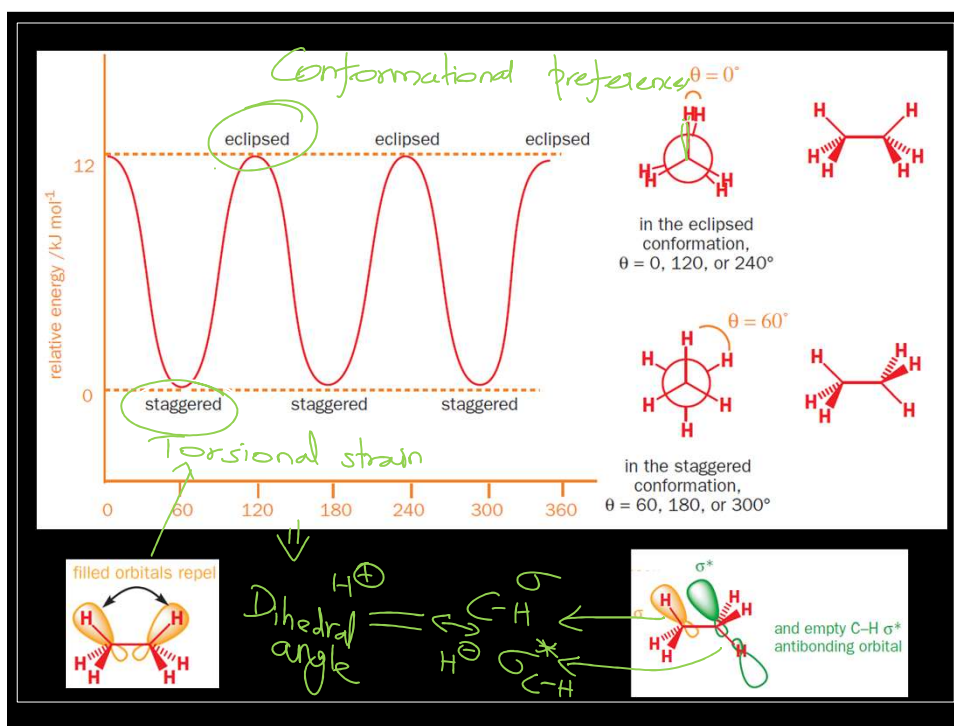
Sawhorse projection

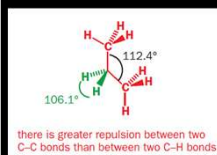


Fisher projection

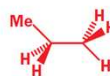


Wedge-dash representation

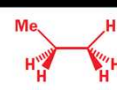




the staggered conformation of propane



the eclipsed conformation of propane



The energy profile diagram for bond rotation in propane would look exactly the same as that for ethane. However, the rotational barrier of propane is slightly higher than for ethane: 14 kJ mol^{-1} as compared to 12 kJ mol^{-1} due to the greater repulsion of electrons in the coplanar bonds in the eclipsed conformation (rather than any steric interactions).

Klyne-Prelog system

